SYSTEM INCIDENT POSTMORTEM

Root Cause Analysis of Antigravity & NotebookLM Subsystem Downtime

Incident Scope: MCP Bridge, Supabase DB & Node Hierarchy	**Severity Level:** High (Service Disruption & Auth Failure)
Investigation Target: Logs, DB Migrations & Infrastructure	**Resolution Status:** Resolved & Architecture Hardened

1. Executive Summary

A multi-layered system degradation and outage occurred across the **Antigravity ecosystem**, affecting both the **Frontend Node Hierarchy Manager** and the **NotebookLM MCP Backend Bridge**. Forensic investigation across server logs (notebooklm.log), CLI crash logs (cookies_debug_v2.log, infographic_run.log), database migrations, and git commit trajectories identified **four primary root causes** that compounded to interrupt operations.

Subsystem	Failure Mode	Direct Technical Trigger	Impact
NotebookLM MCP Bridge	Multi-hop Auth Redirect Loop (401 / 302 to Google Support)	`Expired Google session cookies in auth.json triggering redirect cascade`	CRITICAL
CLI Automation Tools	Uncaught FileNotFoundError	`Missing cookies.txt dependency in invocation paths`	CRITICAL
Supabase DB / Node Tree	Document Insertions Blocked by Constraint Rejection	`Schema added user_id NOT NULL without frontend payload update`	HIGH
Compute / Browser Daemon	Headless Chrome Memory Leaks & OOM Stalls	`High memory consumption across concurrent Selenium sessions`	HIGH

2. Subsystem Root Cause Breakdown

2.1 Google OAuth Session Expiration & Multi-Hop Redirection Loops

The MCP backend (mcp_bp.py) communicates with NotebookLM upstream endpoints. When Google session cookies expired, requests to artifact media URLs (lh3.googleusercontent.com) were rejected and forced through a 5-hop redirect loop:

- **Hops 0 & 1:** 302 Redirect to lh3.google.com/rd-notebooklm/...
- **Hops 2 & 3:** 302 Redirect to accounts.google.com/ServiceLogin (login page)
- **Hops 4 & 5:** 302 Redirect to Google Support (support.google.com/accounts/answer/32050)

The bridge received an HTML document instead of the expected image/audio blob, logging DOWNLOAD FAILED: Received HTML instead of media and returning a 401 Unauthorized error to clients.

```
Log Trace (ManausNotebookLM-baseline/logs/notebooklm.log):
[WARNING] Auth redirect detected mid-chain. Attempting auto-heal...
Hop 4: 302 URL: https://accounts.google.com/ServiceLogin?continue=...
Hop 5: 200 URL: https://support.google.com/accounts/answer/32050?hl=en
[ERROR] DOWNLOAD FAILED: Received HTML instead of media. Content-Type: text/html
werkzeug - INFO - GET /api/mcp/proxy_artifact?url=... HTTP/1.1 401
```

2.2 Missing File Exceptions (`cookies.txt` FileNotFoundError)

Standalone generation scripts (e.g. create_infographic.py, test_cookies_v2.py) assumed cookies.txt was situated in their current execution directory. When run from different working roots or clean environments, scripts crashed instantly: FileNotFoundError: [Errno 2] No such file or directory: 'cookies.txt'.

2.3 Database Schema & RLS Constraint Mismatch (Cascade Delete Migration)

A database migration added a strict foreign key constraint: user_id UUID NOT NULL REFERENCES auth.users(id) ON DELETE CASCADE. Because frontend services (HierarchyService.ts, NodeService.ts) and the create_node RPC function originally omitted the user_id field in insert payloads, Supabase rejected all node creations, causing the visual hierarchy editor to freeze and fail silently on document saves.

2.4 Memory Pressure & Headless Browser Resource Bottlenecks

Headless Chrome / Selenium worker processes spawned by the backend consumed excessive memory over continuous sessions. This led to system thrashing and out-of-memory (OOM) aborts, resolved via git commit 6b331cb ('Swith to 64GB') to scale host memory capacity.

3. Restorative Actions & System Modernization

- **NLM-Everywhere CLI Architecture:** Migrated from fragile browser automation to the official/decoupled nlm CLI (commit 0bf4f95 & 8ceb281), eliminating headless Chrome session crashes.
- **Supabase Schema & RPC Alignment:** Updated HierarchyService.ts, NodeService.ts, and supabase_migration_create_node_function.sql to inject authenticated user_id and enforce Row Level Security (RLS).
- **Database-Backed API Key Security:** Moved API keys from browser localStorage to encrypted Supabase tables (supabase_migration_api_keys.sql) with per-user RLS policies.
- **Host Resource Expansion:** Upgraded host VM RAM to 64GB and configured process recycling to guarantee container stability.

4. Preventative Safeguards & Health Monitoring

Domain	Preventative Measure	Target Safeguard / SLA
Auth Management	Proactive Token Refresh & Expiry Telemetry	Alert 24h prior to Google session expiration
Database Integrity	Automated TypeScript-to-SQL Schema Linter	Pre-commit validation of RPC parameters and RLS policies
Process Reliability	Automated Process Lifecycle Monitoring	Restart idle browser daemons after 15m; cap memory per worker
Path Resilience	Centralized Configuration & Path Resolvers	Explicit fallback paths with actionable error prompts in UI